Probability & Statistics

Unit-IV

Test of Hypothesis

Test of Hypothesis

To reach decisions about populations on the basis of sample information, we make certain assumptions about the populations involved. Such assumptions, which may or may not be true, are called statistical hypothesis. By testing a hypothesis is meant a process for deciding whether to accept or reject the hypothesis.

Null Hypothesis

The hypothesis formulated for the sake of rejecting it, under the assumption that it is true, is called the null hypothesis and is denoted by H_0. To test whether one procedure is better than another, we assume that there is no difference between the procedures. Similarly to test whether there is a relationship between two variates, we take H_0 that there is no relationship. By accepting a null hypothesis, we mean that on the basis of the statistic calculated from the sample, we do not reject the hypothesis.

Alternative Hypothesis

Any hypothesis which contradicts the Null Hypothesis is called an Alternative Hypothesis, usually denoted by H_1. The two hypotheses H_0 and H_1 are such that if one is true, the other is false.

For example, if we want to test the null hypothesis that the population has a specified mean μ_0 (say) i.e.,

Null hypothesis H_0: $\mu = \mu_0$

Then alternative hypothesis would be

Alternative hypothesis H_1: (i) $\mu \neq \mu_0$ (either $\mu > \mu_0$ or $\mu < \mu_0$) or

(ii) $\mu > \mu_0$ or

(iii) $\mu < \mu_0$

The alternative hypothesis (i) is known as a two-tailed alternative and (ii) is known as right-tailed alternative and (iii) is known as left-tailed alternative

Errors

Type I Error

If a hypothesis (H_0) is rejected while it should have been accepted, we say that a Type I error has been committed. It is also called α error.

If we write P(reject H_0 when it is true) = P(Type I error) = α

Then α is called size of type I error and is known as producer's risk.

Type II error

If a hypothesis (H_0) is accepted while it should have been rejected, we say that a Type II error has been made. It is also called β error.

If we write P(accept H_0 when it is wrong) = P(Type II error) = β

Then β is called size of type II error and is known as consumer's risk.
The statistical testing of hypothesis aims at limiting the Type I error to a preassigned value (say: 5% or 1%) and to minimize the Type II error.

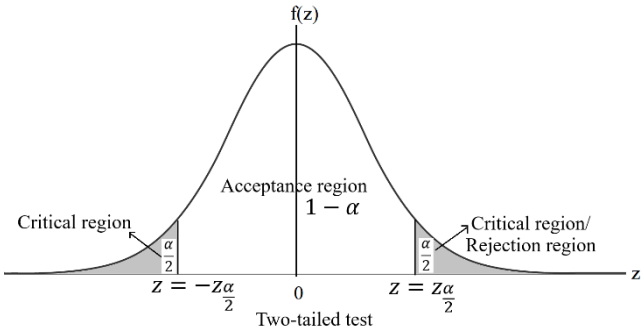
Table of error types		Null hypothesis (H_0) is	
		True	False
Decision about null hypothesis (H_0)	Not reject (Accept)	Correct inference	Type II error
	Reject	Type I error	Correct inference

Level of Significance (α)
The probability level below which we reject the hypothesis is known as the level of significance. The level of significance denoted by α is the confidence with which we reject or accept the null hypothesis H_0 i.e., it is the maximum possible probability with which we are willing to risk an error in rejecting H_0 when it is true. The level of significance is generally specified before a test procedure so that the results obtained may not influence our decision. In practice, we take either 5% (i.e., 0.05) or 1% (i.e., 0.01) level of significance.

The region in which a sample value falling is rejected, is known as critical region. The critical region depends on the test statistic and the type of alternative hypothesis and α. Thus, the probability of the value of the variate falling in the critical region is the level of significance.

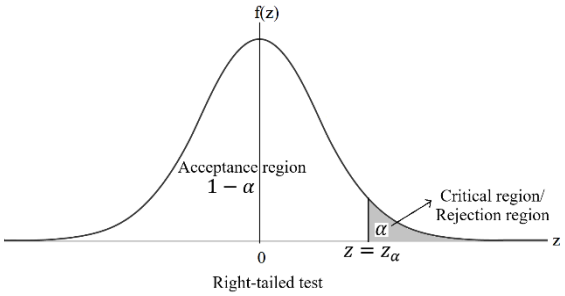
One-tailed and two-tailed tests
Depending on the nature of the problem, we use a single-tailed test or double-tailed test to estimate the significance of a result.

If the alternative hypothesis is of the type (i) $H_1: \mu \neq \mu_0$, then two-tailed test is applied.
In a two-tailed test, the areas of both the tails of curve representing the sampling distribution are taken into account.



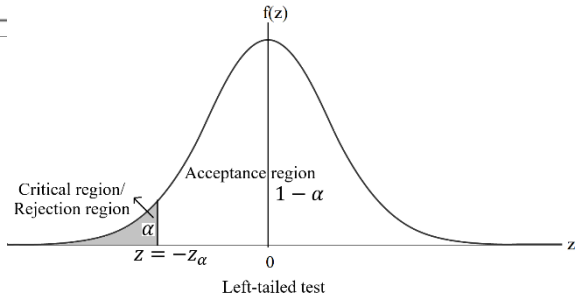
Two-tailed test

If the alternative hypothesis is of the type (ii) $H_1: \mu > \mu_0$ or (iii) $H_1: \mu < \mu_0$ then one-tailed test is applied.
In a one-tailed test, only the area on one side of an ordinate is taken into consideration.
If $H_1: \mu > \mu_0$ then a right tailed test is applied.



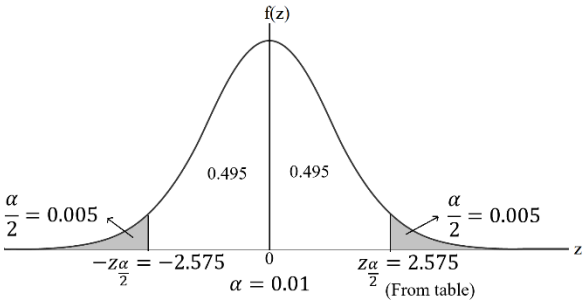
Right-tailed test

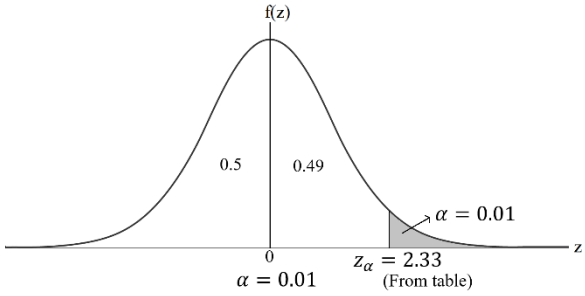
If $H_1: \mu < \mu_0$ then a left tailed test is applied.



Critical values (z_α) of z

Level of significance (α)	Critical values of z		
	1% $(\alpha = 0.01)$	5% $(\alpha = 0.05)$	10% $(\alpha = 0.10)$
Two-tailed test	$\left\|z_{\frac{\alpha}{2}}\right\| = 2.575$	$\left\|z_{\frac{\alpha}{2}}\right\| = 1.96$	$\left\|z_{\frac{\alpha}{2}}\right\| = 1.645$
Right tailed test	$z_\alpha = 2.33$	$z_\alpha = 1.645$	$z_\alpha = 1.28$
Left tailed test	$z_\alpha = -2.33$	$z_\alpha = -1.645$	$z_\alpha = -1.28$





The value of the test statistic which separates the rejection region (critical region) and the acceptance region is called the critical value or significant value. This value is dependent on

(i) the level of significance used, and
(ii) the alternative hypothesis, whether it is one-tailed or two-tailed.

Tests of significance

The procedure which enables us to decide whether to accept or reject the hypothesis is called the test of significance.

Single mean:

Let a random sample of size n $(n \geq 30)$ has the sample mean $\overline{x}$, and μ be the population mean and σ be the population standard deviation. Also, the population mean has a specified value μ_0.

Step 1. Null Hypothesis (H_0): $\mu = \mu_0$.

Step 2. Alternative Hypothesis (H_1): (i) $\mu \neq \mu_0$ (i.e., $\mu > \mu_0$ or $\mu < \mu_0$) (Two-tailed test)

Or (ii) $\mu > \mu_0$ (Right tailed test)

Or (iii) $\mu < \mu_0$ (Left tailed test)

Step 3. Level of Significance (α): Set the level of significance α.

Critical value of z is z_α (one-tailed test) or $z_{\frac{\alpha}{2}}$ (two-tailed test).

Step 4. Test Statistic: $z = \frac{\overline{x}-\mu}{\frac{\sigma}{\sqrt{n}}}$

Step 5. Conclusion: If $|z| < |z_\alpha|$, we accept the null hypothesis H_0.

If $|z| > |z_\alpha|$, we reject the null hypothesis H_0 and accept the alternative hypothesis H_1

1. An oceanographer wants to check whether the depth of the ocean in a certain region is 57.4 fathoms, as had previously been recorded. What can he conclude at 0.05 level of significance, if reading taken at 40 random locations in the given region yielded a mean of 59.1 fathoms with a standard deviation of 5.2 fathoms?

   Sample size: $n = 40$ population mean: $\mu = 57.4$fathoms

   Sample mean: $\bar{x} = 59.1$fathoms

   Sample standard deviation: $s = 5.2$fathoms

   Significance level: $\alpha = 0.05$

   Null hypothesis: $H_0: \mu = \mu_o$

   Alternative hypothesis: $H_1: \mu \neq \mu_o$ (two-tailed test)

   Test Statistic

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

$$Z = \frac{59.1 - 57.4}{5.2/\sqrt{40}}$$

$$|Z| = 2.067$$

Critical Value: $Z_\alpha = 1.96$ from table for $\alpha = 0.05$

Clearly $|Z| > Z_\alpha$

Null hypothesis is Rejected.

2. An ambulance service claims that it takes on an average of less than 10 minutes to reach its destination in emergency calls. A sample of 36 calls has a mean of 11 minutes and the variance of 16 minutes. Test the significance at 0.05 level of significance.

Sample size: $n = 36$ population mean: $\mu = 10$

Sample mean: $\bar{x} = 11$

Sample standard deviation: $s = 4$

Significance level: $\alpha = 0.05$

Null hypothesis: $H_0: \mu = \mu_o$

Alternative hypothesis: $H_1: \mu < \mu_o$ (Left-tailed test)

Test Statistic

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

$$Z = \frac{11 - 10}{4/\sqrt{36}}$$

$|Z| = 1.5$

Critical Value: $Z_\alpha = 1.645$ from table for $\alpha = 0.05$

Clearly $|Z| < Z_\alpha$

Null hypothesis is accepted.

Confidence limits:

$$E_{\max} = Z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} = \frac{1.96*4}{\sqrt{36}} = 1.307$$

$$(\bar{x} - E_{\max}, \bar{x} + E_{\max}) = (11-1.307, 11+1.307)$$

3. A trucking firm suspects the claim that the average life time of certain types is at least 28,000 miles. To check the claim the firm puts 40 of these tyres on its trucks and gets a mean life time of 27,463 miles with a standard deviation of 1348 miles. Check the claim with 0.01 LOS
4. A sample of 100 students have a mean weight of 50kgs. can this be regarded as a sample from population with mean weight 56 kgs with standard deviation 16 kgs. Test at 1 % level of significance.

Difference of means

Test of significance of two means (Large samples, $n \geq 30$)
(Test of significance for difference of means of two large samples)
Let $\bar{x}_1$ and $\bar{x}_2$ be the sample means of two independent large random samples of sizes n_1 and n_2 drawn from two populations having means μ_1 and μ_2 and standard deviations σ_1 and σ_2. To test whether the two-population means are equal or not.
Let a random sample of size n $(n \geq 30)$ has the sample mean $\bar{x}$, and μ be the population mean and σ be the population standard deviation. Also, the population mean has a specified value μ_0.
Step 1. Null Hypothesis (H_0): $\mu_1 = \mu_2$.
Step 2. Alternative Hypothesis (H_1): (i) $\mu_1 \neq \mu_2$ (i.e., $\mu_1 > \mu_2 or \mu_1 < \mu_2$) (Two-tailed test) Or (ii) $\mu_1 > \mu_2$ (Right tailed test)
Or (iii) $\mu_1 < \mu_2$ (Left tailed test)
Step 3. Level of Significance (α): Set the level of significance α.
Critical value of z is z_α (one-tailed test) or $z_{\frac{\alpha}{2}}$ (two-tailed test).
Step 4. Test Statistic: $z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{{\sigma_1}^2}{n_1} + \frac{{\sigma_2}^2}{n_2}}}$
Step 5. Conclusion: If $|z| < |z_\alpha|$, we accept the null hypothesis H_0.
If $|z| > |z_\alpha|$, we reject the null hypothesis H_0 and accept the alternative hypothesis H_1

1. The mean yield of wheat from a district A was 210 pounds with S.D 10 pounds per acre from a sample of 100 plots. In another district the mean yield was 200 pounds with S.D 12 pounds from a sample of 150 plots. Assuming that the S.D of yield in the entire state was 11 pounds test whether there is any significant difference between the mean yields of crops in the two districts.

Sample 1: sample 2:

District A: $\bar{x}_1 = 210, n_1 = 100, s_1 = 10$ District B: $\bar{x}_2 = 200, n_2 = 150, s_2 = 12$

Significance level: $\alpha = 0.05$

Hypotheses

- $H_0: \mu_1 = \mu_2$
- $H_1: \mu_1 \neq \mu_2$ (two-tailed test)

Test Statistic

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Substitute values:

$$Z = \frac{210 - 200}{\sqrt{\frac{10^2}{100} + \frac{12^2}{150}}}$$

$$|Z| = 7.143$$

Critical Value: $Z_\alpha = 1.96$ from table for $\alpha = 0.05$

Clearly $|Z| > Z_\alpha$

Null hypothesis is Rejected.

2. In a survey of buying habits, 400 women shoppers are chosen at random in supermarket 'A' located in a certain section of the city. This average weekly food expenditure is rupees 250 with S.D of rupees 40. For 400 women shoppers chosen at random in supermarket 'B' in another section of the city, the average weekly food

expenditure is rupees 220 with a S.D of rupees 55. Test 1% level of significance whether the average food expenditure of the two populations of shopkeepers are equal.

Sample 1: sample 2:

District A: $\bar{x}_1 = 250, n_1 = 400, s_1 = 40$ District B: $\bar{x}_2 = 220, n_2 = 400, s_2 = 55$

Significance level: $\alpha = 0.01$

Hypotheses

- $H_0: \mu_1 = \mu_2$
- $H_1: \mu_1 \neq \mu_2$ (two-tailed test)

Test Statistic

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Substitute values:

$$Z = \frac{250 - 220}{\sqrt{\frac{40^2}{400} + \frac{55^2}{400}}}$$

$$|Z| = 8.82$$

Critical Value: $Z_\alpha = 2.58$ from table for $\alpha = 0.01$

Clearly $|Z| > Z_\alpha$

Null hypothesis is Rejected.

3. At a certain large University, a sociologist speculates that male students spend considerably more money on junk food than do female students. To test her hypothesis, the sociologist randomly selects from the registrars' records the names of 200 students. Of these 125 are men and 75are women. The average amount spent on junk food per week by the men is Rs.400/- and standard

deviation is Rs100/-. For the women, the sample mean is Rs.450/- and the S.D is Rs.150/-.Test the difference between the mean at 0.05 level of significance.

4. A researcher wants to know the intelligence of students in a school. He selected two groups of students. In the first group there are 150 students having mean IQ of 75 with a standard deviation of 15, in the second group there are 250 students having mean IQ of 70 with standards deviation of 20. Test whether there is any significant difference.
5. A company claims that its light bulbs are superior to those of its main competition. A study showed that a sample of 40 of its bulbs has a mean lifetime of 1647 hours with a standard deviation of 27 hours. Meanwhile, a sample of 40 bulbs from the competitor has a mean lifetime of 1638 hours with a standard deviation of 31 hours. Does this substantiate the claim at the 0.05 level of significance?

Single Proportion

Test of significance for single proportion (Large samples, $n \geq 30$)

Suppose a large random sample of size $n\ (n \geq 30)$ has a sample proportion p of members possessing a certain attribute (i.e., proportion of success). To test the hypothesis that the proportion P in the population has a specified value P_0.

Step 1. Null Hypothesis (H_0): $P = P_0$ (P_0 is a particular value of P)

Step 2. Alternative Hypothesis (H_1): (i) $P \neq P_0$ (i.e., $P > P_0$ or $P < P_0$) (Two-tailed test)

Or (ii) $P > P_0$ (Right tailed test)

Or (iii) $P < P_0$ (Left tailed test)

Step 3. Level of Significance (α): Set the level of significance α.

Critical value of z is z_α (one-tailed test) or $z_{\frac{\alpha}{2}}$ (two-tailed test).

Step 4. Test Statistic: $Z = \frac{p - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$

Step 5. Conclusion: If $|z| < |z_\alpha|$, we accept the null hypothesis H_0.

If $|z| > |z_\alpha|$, we reject the null hypothesis H_0 and accept the alternative hypothesis H_1.

1. A manufacturer claims that at least 95% of the equipment supplied to a factory conforms to specification. A sample of 200 pieces of equipment reveals that 18 are faulty. Test the manufacturer's claim at significance levels 5%.

Sample size: $n = 200$

Faulty pieces: 18

Sample proportion:

$$\mathrm{p} = \frac{182}{200} = 0.91$$

Population proportion: $p_0 = 0.95$

Significance level: $\alpha = 0.05$

Null hypothesis: $H_0: p = p_o$

Alternative hypothesis: $H_1: p \neq p_o$(Two-tailed test)

Test Statistic

The z-test for proportions:

$$Z = \frac{p - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

$$Z = \frac{0.91 - 0.95}{\sqrt{\frac{0.95 * 0.05}{200}}}$$

$$|Z| = 2.59$$

Critical Value: $Z_\alpha = 1.96$ from table for $\alpha = 0.05$

Clearly $|Z| > Z_\alpha$

Null hypothesis is Rejected.

2. In a sample of 1000 people in Karnataka, 540 are rice eaters and the rest are wheat eaters. Can we assume that both rice and wheat are equally popular in this state at a 1% level of significance?

Sample size: $n = 1000$

Rice eaters: 540

Sample proportion:

$$\mathrm{p} = \frac{540}{1000} = 0.54$$

Population proportion: $p_0 = 0.5$

Significance level: $\alpha = 0.01$

Null hypothesis: $H_0: p = p_o$

Alternative hypothesis: $H_1: p \neq p_o$(Two-tailed test)

Test Statistic

The z-test for proportions:

$$Z = \frac{p - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

$$Z = \frac{0.54 - 0.5}{\sqrt{\frac{0.5 * 0.5}{1000}}}$$

$$|Z| = 2.532$$

Critical Value: $Z_\alpha = 2.56$ from table for $\alpha = 0.01$

Clearly $|Z| < Z_\alpha$

Null hypothesis is accepted.

3. In a big city 325 men out of 600 men were found to be smokers. Does this information support the conclusion that the majority of men in this city are smokers?

4. In a random sample of 120 cool drinkers 68 said they prefer Thums up to Pepsi. Test the Null hypothesis p=0.5 against the alternative hypothesis p>0.5.

Difference of Proportions

Let p_1 and p_2 be the sample proportions in two large random samples of sizes n_1 and n_2 drawn from two populations having proportions P_1 and P_2.
To test whether the two samples have been drawn from the same population.
Step 1. Null Hypothesis (H_0): $P_1 = P_2$.
Step 2. Alternative Hypothesis (H_1): $P_1 \neq P_2$ (Two-tailed test)
Step 3. Level of Significance (α): Set the level of significance α.
Critical value of z is Z_α
Step 4. Test Statistic: There are two ways of computing a test statistic z

$$Z = \frac{p_1 - p_2}{\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}}$$

Step 5. Conclusion: If $|z| < Z_\alpha$, we accept the null hypothesis H_0.
If $|z| > Z_\alpha$, we reject the null hypothesis H_0 and accept the alternative hypothesis H_1.

1. A manufacturer of electronic equipment subjects to samples of two completing brands of transistors to an accelerated performance test. If 45 of 180 transistors of the first kind and 34 of 120 transistors of the second kind fail the test. What can we conclude at the level of significance at 5% about the difference between the corresponding sample proportions?

Brand 1: $n_1 = 180$, $x_1 = 45$failures

$P_1 = \frac{45}{180} = 0.25$

Brand $2 n_2 = 120$, $x_2 = 34$failures

$p_2 = \frac{34}{120} \approx 0.2833$

$\alpha = 0.05$.

- Null hypothesis: $H_0: p_1 = p_2$

- Alternative hypothesis: $H_1: p_1 \neq p_2$(two-tailed test)

$$Z = \frac{p_1 - p_2}{\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}}$$

$|Z| = 0.58$

Critical Value: $Z_\alpha = 1.96$ from table for $\alpha = 0.05$

Clearly $|Z| < Z_\alpha$

Null hypothesis is accepted.

2. On the basis of their total scores, 200 candidates of a civil services examination are divided into two groups, the upper 30% and the remaining 70%. Consider the first question of the examination. Among the first group 40 had correct answer whereas the second group 80 had the correct answer. On the basis of these results, can one conclude that the first question is not good at discriminating ability of the type being examined here?

- Total candidates = 200
- Upper 30% group → $n_1 = 60$ candidates
    - Correct answers = 40 → $p_1 = \frac{40}{60} = 0.667$
- Remaining 70% group → $n_2 = 140$candidates
    - Correct answers = 80 → $p_2 = \frac{80}{140} = 0.571$

$\alpha = 0.05$.

- $H_0: p_1 = p_2$
- $H_1: p_1 \neq p_2$(Two tailed test)

$$Z = \frac{p_1 - p_2}{\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}}$$

$|Z| = 1.150$

Critical Value: $Z_\alpha = 1.96$ from table for $\alpha = 0.05$

Clearly $|Z| < Z_\alpha$

Null hypothesis is accepted.

3. In two large populations, there are 30 %and 25 % respectively of fair-haired people. Is this different likely to be hidden in sample of 1200 and 900 resp. From the two population. Evaluate Hypothesis

4. Random samples of 400 men and 600 women were asked whether they would like to have flyover near their residence, 200 men and 325 women were in favor of the proposal. Test the hypothesis that proportions of men and women in favor of the proposal are same against that they are not, at level

5. A coffee company claims that brand A outsells its brand B by 8%. If 42 out 200 prefer brand A and 18 out of 100 prefer brand B. Test the claim at 1% level.

Small Samples

Single Mean (t -test for single mean)

Test of significance for small samples (Small samples, $n < 30$)
Student's t-test for single mean
Step 1. Null Hypothesis (H_0): $\mu = \mu_0$.
Step 2. Alternative Hypothesis (H_1): (i) $\mu \neq \mu_0$ (i.e., $\mu > \mu_0$ or $\mu < \mu_0$) (Two-tailed test)
Or (ii) $\mu > \mu_0$ (Right tailed test)
Or (iii) $\mu < \mu_0$ (Left tailed test)
Step 3. Level of Significance (α): Set the level of significance α.
Critical value of t is t_α (one-tailed) or $t_{\frac{\alpha}{2}}$ (two-tailed) with degree of freedom $\nu = n - 1$.
Step 4. Test Statistic:

$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n-1}}}$, if sample standard deviation(s) is given.

Step 5. Conclusion: If $|t| < |t_\propto|$, we accept the null hypothesis H_0.
If $|t| > |t_\propto|$, we reject the null hypothesis H_0 and accept the alternative hypothesis H_1.

1. A sample of 26 bulbs gives a mean life of 990 hours with a standard deviation of 20 hours. The manufacturer claims that the mean life of bulbs is 1000 hours. Is this sample not up to the standard?

Sample size: $n = 26$ population mean: $\mu = 1000$

Sample mean: $\bar{x} = 990$

Sample standard deviation: $s = 20$

Significance level: $\alpha = 0.05$

Null hypothesis: $H_0: \mu = \mu_o$

Alternative hypothesis: $H_1: \mu < \mu_o$ (left-tailed test)

Test Statistic

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n-1}}$$

$$t = \frac{990 - 1000}{20/\sqrt{26-1}}$$

$|t|$=2.5

Critical Value: $t_\alpha = 1.708$ from table for $\alpha = 0.05, v = n - 1 = 25$

Clearly $|t| > t_\alpha$

Null hypothesis is Rejected.

2. The heights of 10 males of a given locality are found to be 70,67,62,68,61,68,70,64,64,66 inches. Is it reasonable to believe that the average height is greater than 64 inches Test at 5%.

Sample size: $n = 10$ population mean: $\mu = 64$

Compute sample mean

$$\bar{x} = \frac{\text{sum of values}}{n}$$

$$\bar{x} = \frac{660}{10} = 66$$

Compute sample standard deviation

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

$= (70-66)^2 + (67-66)^2 + (62-66)^2 + (68-66)^2\ (61-66)^2 + (68-66)^2 + (70-66)^2 + (64-66)^2 + (64-66)^2 + (66-66)^2/9$

$$s = \sqrt{\frac{86}{9}} = \sqrt{10} \approx 3.16$$

Significance level: $\alpha = 0.05$

Null hypothesis: $H_0: \mu = \mu_o$

Alternative hypothesis: $H_1: \mu > \mu_o$ (Right-tailed test)

Test Statistic

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n-1}}$$

$$t = \frac{66 - 64}{3.16/\sqrt{10-1}}$$

$|t|$=1.90

Critical Value: $t_\alpha = 1.833$ from table for $\alpha = 0.05, v = n - 1 = 9$

Clearly $|t| > t_\alpha$

Null hypothesis is Rejected.

Confidence interval or Confidence limits:

$$E_{\max} = t_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}} = \frac{2.262 * 3.16}{\sqrt{10}} = 2.26$$

$$(\bar{x} - E_{\max}, \bar{x} + E_{\max}) = (66 - 2.26, 66 + 2.26)$$

To find $t_{\frac{\alpha}{2}}$, use two tail test in table

3. Producer of gutka, claims that the average nicotine content in his gutka is 1.83 . Can this claim be accepted if a random sample of 8 gutkas of this type have the nicotine contents of 2.0,1.7,2.1,1.9,2.2,2.1,2.0,1.6mg at 5 %level.

$\mu_0 = 1.83$

Sample size: $n = 8$

Sample values: 2.0,1.7,2.1,1.9,2.2,2.1,2.0,1.6

Significance level: $\alpha = 0.05$

Compute sample mean

$$\bar{x} = \frac{\text{sum of values}}{n}$$

$$\bar{x} = \frac{15.6}{8} = 1.95$$

sample standard deviation

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

$= (2.0 - 1.95)^2 + (1.7 - 1.95)^2 + (2.1 - 1.95)^2 + (1.9 - 1.95)^2\ (2.2 - 1.95)^2 + (2.1 - 1.95)^2 + (2.0 - 1.95)^2 + (1.6 - 1.95)^2/7$

$$s = 0.21$$

Null hypothesis: $H_0: \mu = \mu_o$

Alternative hypothesis: $H_1: \mu \neq \mu_o$ (two-tailed test)

Test Statistic

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n-1}}$$

$$t = \frac{1.95 - 1.83}{0.21/\sqrt{8-1}}$$

$|t|$=1.54

Critical Value: $t_\alpha = 2.365$ from table for $\alpha = 0.05, \nu = 8 - 1 = 7$

Clearly $|t| < t_\alpha$

Null hypothesis is accepted.

4. A certain medicine is given to each of the 9 patients resulted following: 7, 3, -1, 4, -3, 5, 6, -4, -1. Can it be concluded that the medicine will general be accompanied by an increase of blood pressure test at 5% level. Assume mean blood pressure is zero.
5. In a random sample of 10 bolts produced by a machine the mean length of bolt is 0.53 mm and standard deviation 0.03 mm. Can we claim from this that the machine is in proper working order if in the past it produced bolts of length 0.50 mm? Use (i) 0.05 (ii) 0.01 L.O.S.

Difference of means (t -test for Difference of means)

Student's t-test for difference of means

Let $\overline{x}$ and $\overline{y}$ be the means of two independent samples of sizes $n_1(< 30)$ and $n_2(< 30)$ drawn from two normal populations having means μ_1 and μ_2. To test whether the two population means are equal (i.e., to test whether the difference $\mu_1 - \mu_2$ is significant).

Step 1. Null Hypothesis (H_0): $\mu_1 = \mu_2$.

Step 2. Alternative Hypothesis (H_1): (i) $\mu_1 \neq \mu_2$ (i.e., $\mu_1 > \mu_2 or \mu_1 < \mu_2$) (Two-tailed test) Or (ii) $\mu_1 > \mu_2$ (Right tailed test)
Or (iii) $\mu_1 < \mu_2$ (Left tailed test)

Step 3. Level of Significance (α): Set the level of significance α. Critical value of t is t_α (one-tailed) or $t_{\frac{\alpha}{2}}$ (two-tailed) with degrees of freedom $\nu = n_1 + n_2 - 2$.

Step 4. Test Statistic

$$t = \frac{\overline{x}_1 - \overline{x}_2}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

follows t-distribution with $\nu = n_1 + n_2 - 2$ d.f.

where

$$s_p^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}$$

Step 5. Conclusion: If $|t| < |t_\alpha|$, we accept the null hypothesis H_0.
If $|t| > |t_\alpha|$, we reject the null hypothesis H_0 and accept the alternative hypothesis H_1.

1. The mean life of 10 electric motors was found to be 1450 hrs with a S.D. of 423 hrs. A second sample of 17 motors chosen from a different batch showed a mean life of 1280 hrs with a S.D. of 398 hrs. Is there a significant difference between the means of the two samples? Use a 0.01 level of significance.

Sample 1: $n_1 = 10$, $\bar{x}_1 = 1450$ hrs, $s_1 = 423$ hrs

Sample 2: $n_2 = 17$, $\bar{x}_2 = 1280$ hrs, $s_2 = 398$ hrs

Significance Level: $\alpha = 0.01$

Null Hypothesis (H_0): $\mu_1 = \mu_2$

Alternative Hypothesis (H_1): $\mu_1 \neq \mu_2$ (Two-tailed test)

$$s_p^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}$$

$$s_p = \sqrt{179{,}286.32} \approx 423.42 \text{ hrs}$$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$|t|$=1.007

Critical Value: $t_\alpha = 2.787$ from table for $\alpha = 0.01$, $v = n_1 + n_2 - 2$=25

Clearly $|t| < t_\alpha$

Null hypothesis is accepted.

2.To examine the hypothesis that the husbands are more intelligent than the wives, an investigator took a sample of 10 couples and administered them a test which measures the I.Q.. The results are as follows:

Husbands	117	105	97	105	123	109	86	78	103	107
Wives	106	98	87	104	116	95	90	69	108	85

Test the hypothesis with a reasonable test at the level of significance of 0.05.
Solution.
We have

Sample 1: $n_1 = 10$, $\bar{x}_1 = 103$, $s_1 = 13.35$ hrs

Sample 2: $n_2 = 10$, $\bar{x}_2 = 95.8$, $s_2 = 13.66$

Null Hypothesis (H_0): $\mu_1 = \mu_2$

Alternative Hypothesis (H_1): $\mu_1 > \mu_2$ (Right-tailed test)

$$s_p^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}$$

$$s_p = 13.51$$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$|t|$=1.191

Critical Value: $t_\alpha = 1.734$ from table for $\alpha = 0.01$, $v = n_1 + n_2 - 2$=18

Clearly $|t| < t_\alpha$

Null hypothesis is accepted.

3. The blood pressure of 5 women before and after intake of a drug is shown below.

Before	110	120	125	132
After	120	118	125	136

Test whether there is significant change in Blood Pressure at α = 0.05.

Sample 1: $n_1 = 4, \bar{x}_1 = 121.75, s_1 = 9.25$

Sample 2: $n_2 = 4, \bar{x}_2 = 124.75, s_2 = 8.06$

Null Hypothesis (H_0): $\mu_1 = \mu_2$

Alternative Hypothesis (H_1): $\mu_1 \neq \mu_2$ (two-tailed test)

$$s_p^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}$$

$$s_p = 8.67$$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$|t|$=0.49

Critical Value: $t_\alpha = 2.44$ from table for $\alpha = 0.01$, $v = n_1 + n_2 - 2$=6

Clearly $|t| < t_\alpha$

Null hypothesis is accepted.

4. Two random samples with following results. Test whether the differences of means is significant at 1% level

Sample	Size	Mean	S.D
1	6	10	4
2	8	12	5

5.Below are given the gain in weights (in kgs.) of pigs fed on two diets and B.
Gain in weight
Diet A:25,32,30,34,24,14,32,24,30,31,35,25
Diet B: 44, 34, 22, 10, 47, 31, 40, 30, 32, 35, 18, 21, 35, 29, 22
Test, if the two diets differ significantly as regards their effect on increase in weight.

Snedecor's F-test of significance

When testing the significance of the difference of the means of two samples, we assumed that the two samples came from the same population or from populations with equal variances. If the variances of the populations are not equal, a significant difference in the means may arise. Hence, before we apply the t-test for the significance of the difference of two means, we have to test for the equality of population variances using F-test of significance.

Test for equality of two population variances

Let two independent random samples of sizes n_1 and n_2 be drawn from two normal populations.

To test the hypothesis that the two population variances σ_1^2 and σ_2^2 are equal.

Step 1. Null Hypothesis (H_0): $\sigma_1^2 = \sigma_2^2$

Step 2. Alternative Hypothesis (H_1): $\sigma_1^2 \neq \sigma_2^2$

Step 3. Level of Significance (α): Set the level of significance α.

$F_\alpha(\upsilon_1, \upsilon_2)\ or\ F_{cal}$ is the value of F with $\nu_1 = n_1 - 1\ and\ \nu_2 = n_2 - 1$ degrees of freedom such that the area under the F-distribution to the right of F_α is α.

Step 4. Test Statistic: $F = \frac{S_1^2}{S_2^2}\ or\ \frac{S_2^2}{S_1^2}\ or\ \frac{greater\ variance}{smaller\ variance}$

S_1^2 and S_2^2 are the estimates of σ_1^2 and σ_2^2

$S_1^2 = \frac{n_1 s_1^2}{n_1 - 1} = \frac{1}{n_1 - 1}[\sum(x_i - \overline{x})^2]$ and $S_2^2 = \frac{n_2 s_2^2}{n_2 - 1} = \frac{1}{n_2 - 1}[\sum(y_i - \overline{y})^2]$

Step 5. Conclusion: If $F < F_\alpha$, we accept the null hypothesis H_0.

If $F > F_\alpha$, we reject the null hypothesis H_0 and accept the alternative hypothesis H_1.

1: In one sample of 8 observations from a normal population, the sum of the squares of deviations of the sample values from the sample mean is 84.4 and in another sample of 10 observations it was 102.6. Test at 5% level whether the populations have the same variance.

Solution. Given

sample size $n_1 = 8$ (<30, small sample) and

sum of squares of deviations of the sample values from the sample mean = 84.4 i.e., $\sum(x_i - \overline{x})^2 = 84.4$,

sample size $n_2 = 10$ (<30, small sample) and

sum of squares of deviations of the sample values from the sample mean = 102.6 i.e., $\sum(y_i - \overline{y})^2 = 102.6$.

Let σ_1^2 and σ_2^2 be the variances of the two normal populations from which the samples are drawn.

If S_1^2 and S_2^2 are the estimates of σ_1^2 and σ_2^2, then

$S_1^2 = \frac{1}{n_1 - 1}[\sum(x_i - \overline{x})^2] = \frac{1}{7}(84.4) = 12.057$ and

$S_2^2 = \frac{1}{n_2 - 1}\left[\sum(y_i - \overline{y})^2\right] = \frac{1}{9}(102.6) = 11.4$

Step 1. Null Hypothesis (H_0): $\sigma_1^2 = \sigma_2^2$

Step 2. Alternative Hypothesis (H_1): $\sigma_1^2 \neq \sigma_2^2$
Step 3. Level of Significance (α): $\alpha = 0.05$.
Step 4. Test Statistic: $F = \frac{S_1^2}{S_2^2}$ since $S_1^2 > S_2^2$

$$F = \frac{S_1^2}{S_2^2} = \frac{12.057}{11.4} = 1.057$$

$F_\alpha(v_1, v_2) = 3.29$ with $v_1 = n_1 - 1 = 8 - 1 = 7 and\ v_2 = n_2 - 1 = 10 - 1 = 9$ degrees of freedom

Step 5. Conclusion: If $F < F_\alpha$, we accept the null hypothesis H_0

2. It is known that the mean diameters of rivets produced by two firms A and B are practically the same, but the standard deviations may differ. For 22 rivets produced by firm A, the S.D is 2.9mm, while for 16 rivets manufactured by firm B, the S.D IS 3.8 mm. compute the statistic you would use to test whether the products of firm A have the same variability as those of firm B and test its significance.

sample size $n_1 = 22$ (<30, small sample) and
sample size $n_2 = 16$ (<30, small sample) and
Let σ_1^2 and σ_2^2 be the variances of the two normal populations from which the samples are drawn.
If S_1^2 and S_2^2 are the estimates of σ_1^2 and σ_2^2, then

$s_1 = 2.9$, $s_2 = 3.8$.
Step 1. Null Hypothesis (H_0): $\sigma_1^2 = \sigma_2^2$
Step 2. Alternative Hypothesis (H_1): $\sigma_1^2 \neq \sigma_2^2$
Step 3. Level of Significance (α): $\alpha = 0.05$.
Step 4. Test Statistic: $F = \frac{S_2^2}{S_1^2}$ since $S_2^2 > S_1^2$

$$F = 14.44/8.41 \approx 1.72$$

$F_\alpha(v_1, v_2) = 2.1508$ with $v_1 = n_1 - 1 = 16 - 1 = 15 and\ v_2 = n_2 - 1 = 22 - 1 = 21$ degrees of freedom
Step 5. Conclusion: If $F < F_\alpha$, we accept the null hypothesis H_0

Example 2: Time taken by the workers in performing a job by method I and method II is given below:

Method I	20	16	26	27	23	22	---
Method II	27	33	42	35	32	34	38

Do the date show that the variances of time distribution from population from which these samples are drawn do not differ significantly?

Solution. Given
sample size $n_1 = 6$ (<30, small sample) and

sample mean $\overline{x} = \frac{\sum x}{n_1} = \frac{20+16+26+27+23+22}{6} = 22.3$
sample size $n_2 = 7$ (<30, small sample) and
sample mean $\overline{y} = \frac{\sum y}{n_2} = \frac{27+33+42+35+32+34+38}{7} = 34.4$

x	$x-\overline{x}$	$(x-\overline{x})^2$	y	$y-\overline{y}$	$(y-\overline{y})^2$
20	-2.3	5.29	27	-7.4	54.76
16	-6.3	39.69	33	-1.4	1.96
26	3.7	13.69	42	7.6	57.76
27	4.7	22.09	35	0.6	0.36
23	0.7	0.49	32	-2.4	5.76
22	-0.3	0.09	34	-0.4	0.16
			38	3.6	12.96
		$\sum(x-\overline{x})^2$ =81.34			$\sum(y-\overline{y})^2$ =133.72

Let σ_1^2 and σ_2^2 be the variances of the two normal populations from which the samples are drawn.
If S_1^2 and S_2^2 are the estimates of σ_1^2 and σ_2^2, then
$S_1^2 = \frac{1}{n_1-1}[\sum(x_i-\overline{x})^2] = \frac{1}{5}(81.34) = 16.26$ and
$S_2^2 = \frac{1}{n_2-1}\left[\sum(y_i-\overline{y})^2\right] = \frac{1}{6}(133.72) = 22.29$
Step 1. Null Hypothesis (H_0): $\sigma_1^2 = \sigma_2^2$
Step 2. Alternative Hypothesis (H_1): $\sigma_1^2 \neq \sigma_2^2$
Step 3. Level of Significance (α): $\alpha = 0.05$. Critical value is
Step 4. Test Statistic: $F = \frac{S_2^2}{S_1^2}$ since $S_2^2 > S_1^2$

$$F = \frac{S_2^2}{S_1^2} = \frac{22.29}{16.26} = 1.37$$

$F_\alpha(v_1, v_2) = 4.39$ with $v_1 = n_1 - 1 = 6 - 1 = 5$ *and* $v_2 = n_2 - 1 = 7 - 1 = 6$ degrees of freedom

$F < F_\alpha$

Step 5. Conclusion: If $F < F_\alpha$, we accept the null hypothesis H_0 and conclude that the populations have the same variance.

Chi-Square (χ^2) test

The magnitude of discrepancy between the theory and observation is given by the quantity χ^2. If $\chi = 0$, the observed and expected frequencies completely coincide. As χ^2 increases, the discrepancy between the observed and theoretical frequencies increases.

If $O_i (i = 1,2, \dots, n)$ is a set of observed (experimental) frequencies and $E_i (i = 1,2, \dots, n)$ is the corresponding set of expected (theoretical) frequencies, then χ^2 is defined as $\chi^2 = \sum_{i=1}^{n} \frac{(O_i - E_i)^2}{E_i}$ with $(n-1)$ degrees of freedom.

χ^2 is used to test whether differences between observed and expected frequencies are significant or not.

Note: If the data is given in a series of $'n'$ numbers, then degrees of freedom = $n - 1$.

In case of Binomial distribution, d.f. = $n - 1$.

In case of Poisson distribution, d.f. = $n - 2$.

In case of Normal distribution, d.f. = $n - 2$.

Chi-square distribution is an important continuous probability distribution and it is used in both large and small samples. In chi-square tests, χ^2-distribution is mainly used

(i) to test the goodness of fit.

(ii) to test the independence of attributes.

(iii) to test if the population has a specified value of the variance σ^2.

χ^2 −test as a test of goodness of fit

We use this test to decide whether the discrepancy between theory and experiment is significant or not.

Step 1. Null Hypothesis (H_0): There is no significant difference between observed values and the corresponding expected values. Assuming the H_0 is true, the test statistic χ^2 follows Chi-square distribution with $(n-1)$ d.f.

Step 2. Alternative Hypothesis (H_1): There is significant difference between observed values and expected values.

Step 3. Level of Significance (α): Set the level of significance α. Critical value of χ^2 is χ_α^2 with degrees of freedom $\nu = n - 1$.

Step 4. Test Statistic: $\chi^2 = \sum_{i=1}^{n} \frac{(O_i - E_i)^2}{E_i}$

Step 5. Conclusion: If $\chi^2 < \chi_\alpha^2$, we accept the null hypothesis H_0.

If $\chi^2 > \chi_\alpha^2$, we reject the null hypothesis H_0 and accept the alternative hypothesis H_1.

1. A sample analysis of examination results of 500 students was made. It was found that 220 students had failed, 170 had secured a third class, 90 were placed in second class and 20 got a first class. Do These figures commensurate with the general examination result which is in the ratio of 4:3:2:1 for the various categories respectively.

Category	Observed (O)	Expected (E)	$O - E$	$(O - E)^2$	$\frac{(O - E)^2}{E}$
Fail	220	200	20	400	2.00
Third class	170	150	20	400	2.67
Second class	90	100	-10	100	1.00
First class	20	50	-30	900	18.00
Total	500	500	–	–	23.67

$\chi 2$=23.67

Degrees of freedom

$$df = k - 1 = 4 - 1 = 3$$

Critical value
At $\alpha = 0.05$, $\chi^2_{\alpha} \approx 7.815$.
Step 6: Decision
$\chi 2 > \chi^2_{\alpha}$

reject null hypothesis.

2. 200 digits are chosen at random from a set of tables. The frequency of the different digits is as follows. Use Chi - square test to assess the hypothesis that the digits are distributed in equal number in the table.

Digit	0	1	2	3	4	5	6	7	8	9
Frequency	18	19	23	21	16	25	22	20	21	15

Digit	Observed (O)	Expected (E)	O – E	(O – E)²	(O – E)² / E
0	18	20	-2	4	0.20
1	19	20	-1	1	0.05
2	23	20	+3	9	0.45
3	21	20	+1	1	0.05
4	16	20	-4	16	0.80
5	25	20	+5	25	1.25
6	22	20	+2	4	0.20

Digit	Observed (O)	Expected (E)	O – E	(O – E)²	(O – E)² / E
7	20	20	0	0	0.00
8	21	20	+1	1	0.05
9	15	20	-5	25	1.25
Total	200	200	—	86	4.30

$\chi 2=4.30$

Degrees of freedom

$$df = k - 1 = 10 - 1 = 9$$

Critical value
At $\alpha = 0.05$, $\chi^2_{\alpha} \approx 16.919$.
Step 6: Decision
$\chi 2 < \chi^2_{\alpha}$
accept null hypothesis.

3. A die is thrown 264 times with the following results. Show that the die is biased.

Number appeared on the die	1	2	3	4	5	6
Frequency	40	32	28	58	54	52

Solution.
The expected frequency of each of the numbers 1, 2, 3, 4, 5, 6 is $\frac{264}{6} = 44$

Calculation of χ^2

Observed frequencies (O_i)	Expected frequencies (E_i)	$(O_i - E_i)^2$	$\frac{(O_i - E_i)^2}{E_i}$
40	44	16	0.3636
32	44	144	3.2727
28	44	256	5.8181
58	44	196	4.4545
54	44	100	2.2727
52	44	64	1.4545
264	264		17.6362

Step 1. Null Hypothesis (H_0): The die is unbiased
Step 2. Alternative Hypothesis (H_1): The die is biased
Step 3. Level of Significance (α): $\alpha = 0.05$. Critical value is
$\chi^2_{\alpha} = \chi^2_{0.05} = 11.070$ with $\nu = n - 1 = 6 - 1 = 5$ degrees of freedom
Step 4. Test Statistic: $\chi^2 = \sum_{i=1}^{n} \frac{(O_i - E_i)^2}{E_i} = 17.6362$
$\therefore \chi^2 > \chi^2_{\alpha}$
Step 5. Conclusion: Since $\chi^2 > \chi^2_{\alpha}$, we reject the null hypothesis H_0 and accept the alternative hypothesis H_1 and conclude that the die is biased.

4. A survey of 240 families, each with 4 children, revealed the following distribution of male births:

Number of male births (x)	4	3	2	1	0
Observed frequency (O)	10	55	105	58	12

Fit a Binomial distribution to the above data and test its goodness of fit at the 5% level of significance.

$$N = 240$$

$$\overline{Mean} = \frac{473}{240} = 1.97$$

For Binomial distribution:

$$Mean = np \Rightarrow 1.97 = 4p \Rightarrow p = 0.4925,\ q = 1 - p = 0.5075$$

- For $x = 4$:

$$P(4) = 4C4 \cdot (0.4925)^4 \cdot (0.5075)^0 = 0.058$$

Expected = $240 \cdot 0.058 = 13.9$

Expected = $240 \cdot 0.374 = 89.8$

- For $x = 3$:

$$P(3) = 4C3 \cdot (0.4925)^3 \cdot (0.5075)^1 = 0.220$$

- For $x = 2$:

$$P(2) = 4C2 \cdot (0.4925)^2 \cdot (0.5075)^2 = 0.374$$

Expected = $240 \cdot 0.220 = 52.8$

- For $x = 1$:

$$P(1) = 4C1 \cdot (0.4925)^1 \cdot (0.5075)^3 = 0.258$$

Expected = $240 \cdot 0.258 = 61.9$

- For $x = 0$:

$$P(0) = 4C0 \cdot (0.4925)^0 \cdot (0.5075)^4 = 0.0666$$

Expected = $240 \cdot 0.0666 = 16.0$

x	Observed (O)	Expected	O – E	$(O-E)^2$	$(O-E)^2$ / Expected
4	10	13.9	-3.9	15.2	1.09
3	55	52.8	+2.2	4.8	0.09
2	105	89.8	+15.2	231.0	2.57
1	58	61.9	-3.9	15.2	0.25
0	12	16.0	-4.0	16.0	1.00
Total	240	234.4 (≈240)	—	—	4.99

$$\chi^2 = 4.99$$

Degrees of freedom:

$$df = \text{categories} - 1 - \text{parameters estimated} = 5 - 1 - 1 = 3$$

Critical value at 5% level, $df = 3$:

$$\chi^2_{\propto} = 7.815$$

$$\therefore \chi^2 < \chi^2_{\alpha}$$

Conclusion: Since $\chi^2 < \chi^2_{\alpha}$, we accept the null hypothesis.

5. Fit a Poisson distribution to the following data and test for its goodness of fit at level of significance 0.05.

x	0	1	2	3	4
f	419	352	154	56	19

Solution.

Total tosses (trials):

$$N = 1000$$

Mean, $\lambda = \frac{\sum f_i x_i}{\sum f_i} = \frac{419(0)+352(1)+154(2)+56(3)+19(4)}{419+352+154+56+19} = \frac{904}{1000} = 0.904$

Expected Frequencies

Poisson probability:

$$P(x) = \frac{e^{-\lambda}\lambda^x}{x!}, \text{ Expected} = N \cdot P(x)$$

- For $x = 0$:

$$P(0) = e^{-0.904} \approx 0.4049$$

Expected = $1000 \cdot 0.4049 = 405$

- For $x = 1$:

$$P(1) = e^{-0.904} \cdot 0.904 \approx 0.366$$

Expected = $1000 \cdot 0.366 = 366$

- For $x = 2$:

$$P(2) = e^{-0.904} \cdot \frac{0.904^2}{2} \approx 0.165$$

Expected = $1000 \cdot 0.165 = 165$

- For $x = 3$:

$$P(3) = e^{-0.904} \cdot \frac{0.904^3}{6} \approx 0.0497$$

Expected = $1000 \cdot 0.0497 = 50$

- For $x = 4$:

$$P(4) = e^{-0.904} \cdot \frac{0.904^4}{24} \approx 0.0112$$

Calculation of χ^2

Observed frequencies (O_i)	Expected frequencies (E_i)	$(O_i - E_i)^2$	$\frac{(O_i - E_i)^2}{E_i}$
419	406.1	166.4	0.4098
352	366.1	198.8	0.5430
154	165.5	132.3	0.7994
56	49.9	37.2	0.7455
19	12.4	43.56	3.5129
			6.0106

Step 1. Null Hypothesis (H_0): Poisson distribution is good fit
Step 2. Alternative Hypothesis (H_1): Poisson distribution is not good fit
Step 3. Level of Significance (α): $\alpha = 0.05$. Critical value is
Step 4. Test Statistic: $\chi^2 = 6.0106$

$\chi^2_\alpha = 7.815$ with $\nu = n - 2 = 5 - 2 = 3$ degrees of freedom

$\therefore \chi^2 < \chi^2_\alpha$

Step 5. Conclusion: Since $\chi^2 < \chi^2_\alpha$, we accept the null hypothesis.

6. Fit a binomial distribution to the following data and test for its goodness of fit at level of significance 0.05.

Number of heads	0	1	2	3	4	5
Number of tosses (Frequency)	38	144	342	287	164	25

Total tosses (trials):

$$N = 1000$$

$$Mean = \frac{2470}{1000} = 2.47$$

Since mean = np:

$$2.47 = 5p \Rightarrow p = 0.494$$

q=1−p=0.506

Formula:

$$P(r) = nCr \cdot p^r \cdot q^{n-r},$$

- For $x = 0$:

$$P(0) = 5C0 \cdot (0.494)^0 \cdot (0.506)^5 = 0.0331$$

Expected = $1000 \cdot 0.0331 = 33.1$

- For $x = 1$:

$$P(1) = 5C1 \cdot (0.494)^1 \cdot (0.506)^4 = 0.161$$

Expected = $1000 \cdot 0.161 = 161.0$

- For $x = 2$:

$$P(2) = 5C2 \cdot (0.494)^2 \cdot (0.506)^3 = 0.312$$

Expected = $1000 \cdot 0.312 = 312.0$

- For $x = 3$:

$$P(3) = 5C3 \cdot (0.494)^3 \cdot (0.506)^2 = 0.305$$

Expected = $1000 \cdot 0.305 = 305.0$

- For $x = 4$:

$$P(4) = 5C4 \cdot (0.494)^4 \cdot (0.506)^1 = 0.149$$

Expected = $1000 \cdot 0.149 = 149.0$

- For $x = 5$:

$$P(5) = 5C5 \cdot (0.494)^5 \cdot (0.506)^0 = 0.0289$$

Expected = $1000 \cdot 0.0289 = 28.9$

x	Observed (O)	Expected	O – E	$(O - E)^2$	$\frac{(O_i - E_i)^2}{E_i}$
0	38	33.1	+4.9	24.0	0.73
1	144	161.0	-17.0	289.0	1.79
2	342	312.0	+30.0	900.0	2.88
3	287	305.0	-18.0	324.0	1.06
4	164	149.0	+15.0	225.0	1.51
5	25	28.9	-3.9	15.2	0.53
Total	1000	989.0 (≈1000)	—	—	8.50

$$\chi^2 = 8.50$$

Degrees of freedom:

$$df = \text{categories} - 1 - \text{parameters estimated} = 6 - 1 - 1 = 4$$

Critical value at 5% level, $df = 4$:

$$\chi^2_{\propto} = 9.488$$

$$\therefore \chi^2 < \chi^2_{\alpha}$$

Conclusion: Since $\chi^2 < \chi^2_{\alpha}$, we accept the null hypothesis.